Yuqing (Yooki) Yang

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Education

University of Toronto GPA: 3.83/4.00 — **Sep 2024 – Oct 2025**
Master of Engineering, Emphasis in Data Analytics and Machine Learning — Toronto, Ontario

York University — **Sep 2017 - Feb 2021**
Honors Bachelor of Science, Computer Science — Toronto, Ontario

Skills

- **Programming & Data Analysis**: Python (Pandas, NumPy, scikit-learn, Matplotlib, Seaborn), SQL, Excel (Solver, VBA, Pivot Tables)
- **Machine Learning & AI**: Supervised Learning (Classification, Regression), Unsupervised Learning (Clustering, PCA), Model Validation, Feature Engineering
- **Big Data & Cloud**: Apache Spark, Hadoop, MapReduce, Azure
- **Development & Collaboration**: Git, Jira
- **Statistical & Financial Methods**: Time Series Analysis (ARIMA, GARCH, Holt-Winters), Monte Carlo Simulation, Risk Analytics (VaR, CVaR), Portfolio Optimization
- **Optimization & Operations**: Linear Programming (CPLEX, Gurobi), Queuing Theory, Process Optimization, Resource Allocation

Work Experience

Financial Advisor Assistant — Sep 2024 – Present

Fountain Wealth Management Inc. | Toronto, ON

- Provided sales support for financial advisor's insurance business, conducting cold calls and successfully scheduling appointments with over 10 prospective clients monthly
- Gathered client information including income, investment status, physical health, and insurance questions, translating client concerns into structured insights for senior consultations
- Collaborated with senior to develop customer-centric insurance solutions, providing data-driven perspectives during client objection handling
- Maintained organized records of client interactions and follow-up schedules to ensure consistent service delivery

Chief Presiding Officer — Apr 2025 & Aug 2025

University of Toronto, Faculty of Arts & Science| Toronto, ON

- Led pre-exam briefings with professors, TAs, and invigilators to communicate proctoring procedures, exam parameters, and regulatory requirements across 40+ exam sessions, ensuring clear understanding and alignment among all stakeholders
- Managed real-time operational incidents including missing exam materials, student medical emergencies, credential verification issues, and room conflicts, demonstrating proactive problem-solving under strict time constraints
- Coordinated exam operations staff of varying team sizes across different examination periods, ensuring efficient resource allocation and seamless execution of examination protocols
- Completed comprehensive incident documentation through multiple reporting systems (general forms and emergency-specific forms) with precise records for compliance and operational continuity

Projects

Portfolio Optimization & Risk Management — Winter 2025

- Built portfolio optimization framework implementing 9 investment strategies (mean-variance, max Sharpe, equal risk contributions, benchmark tracking, leveraged max Sharpe, robust optimization) using linear and quadratic programming solvers
- Tracked risk metrics (Sharpe ratios, variance, max drawdowns) across 2008-2009 financial crisis, 2020-2021 recovery, and 2022 downturn, with benchmark tracking achieving Sharpe 1.91 and drawdown -0.303 in strong market conditions
- Quantified trading frequency differences across strategies (Robust Optimization: 206 trades, Min Variance: 120, Max Sharpe: 86), demonstrating how strategy selection impacts transaction costs and operational execution complexity

Healthcare Operations Optimization — Winter 2025

- Built mixed-integer linear programming model to optimize surgical scheduling for 30 patients across 2 operating rooms and 5 weekdays, managing operational constraints including OR capacity (8am-3pm), overtime costs, and cancellation penalties
- Reduced total operational costs by 50% (from $3,100 to $1,550 in overtime charges) through optimized patient-to-OR-to-day assignments while eliminating cancellations
- Managed data for 30 patient records including OR time requirements (142-239 minutes per patient), prioritizing scheduling decisions to maximize patient throughput while minimizing costs
- Developed clear operational schedule documentation demonstrating resource allocation efficiency and compliance with hospital capacity constraints